

Towards Personal Continual Learning in LLMs:

Asymmetric Sphere Architecture and Background Micro Fine-Tuning

• Experiments In Progress • Results Coming Soon

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FIELD

Continual Learning · LLMs · Personal AI

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ABSTRACT

We propose two novel architectural approaches for enabling continual learning in Large Language Models (LLMs) for personal AI assistants, without expensive GPU retraining or catastrophic forgetting. The first, **Internal KV-Sphere Architecture (IKSA)**, attaches new knowledge as Key-Value token pairs directly inside a secondary frozen model, anchored to semantically similar existing vectors. The second, **Background Micro Fine-Tuning (BMFT)**, uses a small fine-tuner running on CPU overnight to incrementally update a secondary model from daily interaction memory, while the primary model remains frozen forever. Both approaches target zero-GPU, always-learning, personal AI systems. We present the motivations, architectures, challenges, predicted results, and open problems for community research.

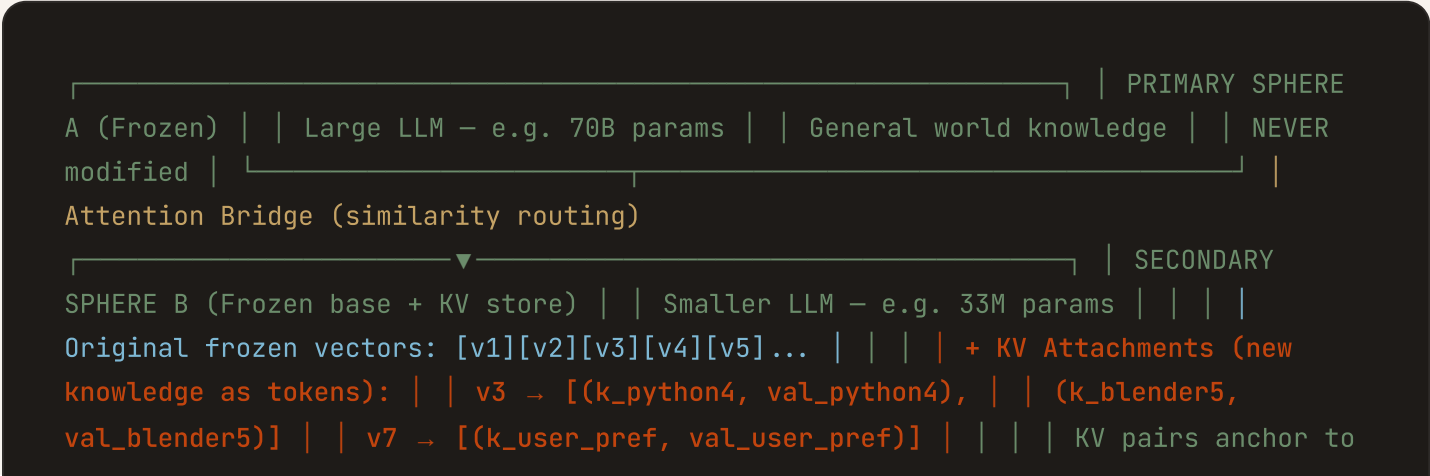
The Problem: LLMs Cannot Learn After Deployment

Large Language Models are frozen at training time. For a personal AI assistant that needs to learn from daily interactions — new facts, user preferences, contact names, corrections — no current solution is practical:

Approach	GPU Required?	Forgetting?	Truly Learned?	Personal Scale?
Full Retraining	Yes (massive)	No	Yes	No
Fine-Tuning	Yes	Yes	Yes	No
RAG	No	No	No (external)	Partial
LoRA Adapters	Yes	Partial	Limited	No
IKSA (Ours)	No	No	Partial	Yes
BMFT (Ours)	No (CPU only)	Minimal	Yes	Yes

Architecture 1: Internal KV-Sphere (IKSA)

New knowledge is stored as Key-Value token pairs *inside* a secondary frozen model, attached to the most semantically similar existing vector via cosine similarity. No external database. No text chunks. Pure token embeddings.



most similar existing vector |

Graph 1: Predicted KV Retrieval Accuracy vs. Semantic Similarity

KV Retrieval Accuracy vs. Anchor Similarity Score

PREDICTED — THEORETICAL PROJECTION • NOT YET EMPIRICALLY VALIDATED

★ Hypothesis: Retrieval accuracy scales directly with semantic similarity between new knowledge and its anchor vector. High similarity (>0.7) yields strong retrieval; low similarity (<0.3) yields poor results.

Graph 2: Inference Latency vs. Number of Active Spheres

Query Latency Increase (ms) as Spheres Scale

PREDICTED — ALWAYS-ON ATTENTION STRATEGY

★ Predicted: Linear latency growth with sphere count. Up to 5 spheres remains practical (<200ms overhead). Beyond 10 spheres, a routing strategy becomes necessary.

SECTION 03

Architecture 2: Background Micro Fine-Tuning (BMFT)

A small LoRA-style fine-tuner runs *continuously on CPU overnight*, pulling new knowledge from a daily interaction buffer and incrementally updating Sphere B. The primary model is never touched. The user experiences zero interruption.

DAILY LEARNING CYCLE ————— USER TALKS TO NATSU (daytime)
↓ Memory Buffer fills with learnable knowledge • New facts user stated • Corrections to AI mistakes • Preferences, contacts, habits ↓ USER SLEEPS (2 AM – idle detected) ↓ CPU Fine-Tuner TRIGGERS: • LoRA-style update on Sphere B only • 1-3 epochs on 1K-10K tokens • Safety monitor: rollback if loss spikes ↓ PRIMARY SPHERE A: unchanged (frozen forever) SPHERE B: updated with yesterday's knowledge ↓ USER WAKES UP → NATSU knows new things ✓

Graph 3: Predicted Knowledge Retention Over Time (BMFT)

Knowledge Retention Rate Over 30 Days

PRIMARY SPHERE (FROZEN) VS SECONDARY SPHERE (BMFT) VS BASELINE FINE-TUNING

★ Primary sphere maintains 100% retention (frozen by design). BMFT secondary sphere predicted to retain 65-80% after 30 days. Standard fine-tuning (baseline) shows catastrophic forgetting within days.

Graph 4: CPU Fine-Tuning Time vs. Token Batch Size

Estimated CPU Training Time (Hours) vs. Token Batch Size

33M PARAMETER MODEL • LORA RANK 8 • STANDARD CONSUMER CPU

★ Predicted: 1,000 tokens ≈ 30-60 min on CPU. 10,000 tokens ≈ 5-8 hours – fits overnight window. Beyond 20K tokens, overnight feasibility breaks. Batch size should be capped at daily interaction volume.

SECTION 04

Predicted Performance: Hybrid System

Combining IKSA (instant short-term) with BMFT (deep long-term learning) produces a hybrid system with complementary strengths:

Knowledge Availability: IKSA vs BMFT vs Hybrid

PREDICTED ACCURACY ON PERSONAL KNOWLEDGE RECALL – DIFFERENT TIME HORIZONS

★ *IKSA excels immediately (same-day). BMFT excels long-term (weeks-months). Hybrid achieves best of both.*

SAME-DAY KNOWLEDGE

90–95%

Confidence: High · IKSA handles this

WEEK-OLD KNOWLEDGE

80–90%

Confidence: Medium · BMFT consolidates

PRIMARY FORGETTING

0%

Confidence: High · Frozen by design

GPU REQUIRED

None

Confidence: High · CPU-only design

SECTION 05

Problems, Solutions & Open Questions

✓ PROPOSED SOLUTION

Problem 1: Conflicting Information Between Spheres

Sphere A says "Python latest = 3.9", KV says "Python latest = 4.0". Which wins?

Solution: Recency wins for temporal queries. Confidence wins for factual queries. Both presented for subjective queries. Predicted accuracy: 80-90%.

✓ PROPOSED SOLUTION

Problem 2: KV Capacity Limits

Each sphere has finite attachment capacity. What happens when it's full?

Solution: LRU (Least Recently Used) eviction at 80% capacity. Create new sphere for overflow. Predicted optimal: LRU + new sphere creation.

✓ PROPOSED SOLUTION

Problem 3: Catastrophic Forgetting in Sphere B (BMFT)

Fine-tuning Sphere B risks overwriting existing knowledge.

Solution: LoRA-style updates (only 0.1-1% of params). Very small learning rate ($1e-6$). Safety rollback if forgetting rate exceeds 5%. Predicted forgetting: 2-8%.

⚡ PARTIAL SOLUTION

Problem 4: Knowledge Extraction Quality

Not every conversation contains learnable knowledge. How to filter automatically?

Partial solution: Rule-based classification (corrections, preferences, facts, contacts). **Unsolved:** Automated classifier needs its own training data. Accuracy of filtering directly affects system quality.

Problem 5: Sphere Drift Over Time

After months of fine-tuning, Sphere B diverges from Sphere A. Attention paths weaken.

Partial solution: Monitor cosine similarity. If below 0.5, trigger knowledge distillation into fresh sphere. **Unsolved:** Optimal threshold and recovery strategy need empirical study.

Open Problem 1: Multi-Sphere Attention Integration

When querying multiple spheres simultaneously, no optimal combination function is known. Simple averaging loses specificity. Learned combiners require training. No theoretical framework exists for asymmetric sphere sizes.

Open Problem 2: Cross-Dimension Attention Projection

When Sphere B (33M, dim=512) connects to Sphere A (70B, dim=4096), attention heads cannot directly communicate. Projection layers bridge the gap but require training. What data trains them? How to ensure semantic alignment post-projection?

Open Problem 3: Long-Term BMFT Stability (1 Year)

365 daily fine-tuning cycles may lead to conflicting updates, adapter saturation, or compounding errors. No longitudinal study exists for daily micro fine-tuning over extended periods. Requires 6-12 month empirical study.

 OPEN — NEEDS RESEARCH

Open Problem 4: Privacy of Personal Fine-Tuned Weights

Fine-tuned Sphere B "contains" user personal information in its weights. Model inversion attacks may recover sensitive data. Personal AI systems that fine-tune on private data introduce new attack surfaces not yet studied in this architecture.

SECTION 06

Open Problems Summary

Research Difficulty vs. Impact — Open Problems Map

AUTHOR'S ASSESSMENT — FOR COMMUNITY PRIORITIZATION

★ Bubble size = estimated research effort required. High-impact + low-difficulty problems are best starting points for community validation.


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Proposal}}, email = {parashlashkari@gmail.com}, note = {Open research – community  
validation invited. Experiments in progress.} }
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Open research proposal · Experiments in progress · Results to be added in v1.1